

AWS Networking Fundamentals



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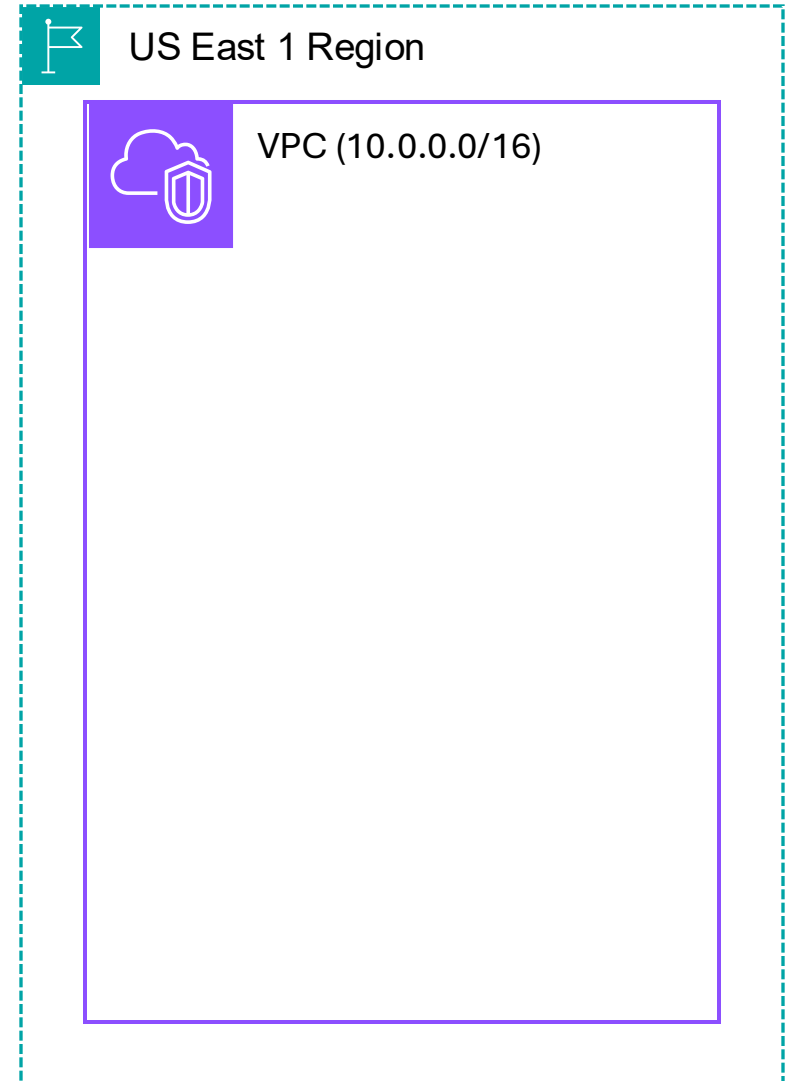


AWS Networking Building Blocks

- Virtual Private Cloud (VPC)
- Subnet
- Route Table
- Internet Gateway
- NAT Gateway
- Network Access Control List (NACL)
- Security Group

Virtual Private Cloud (VPC)

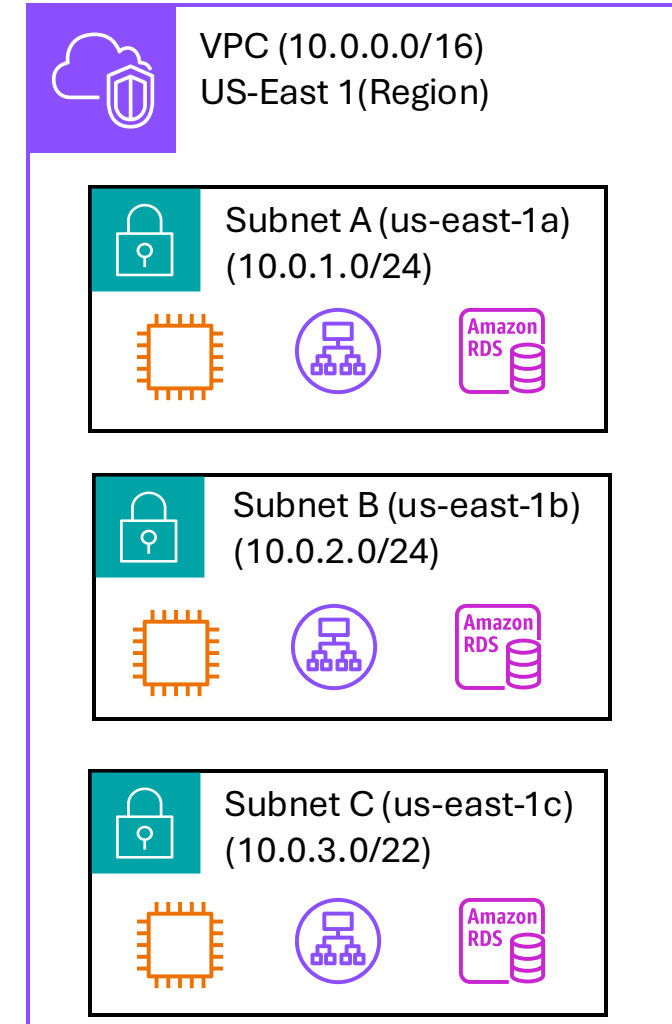
- AWS networking begins with a VPC
- Your private virtual network in the cloud
- Control IP ranges, subnets, and routing
- Isolated and secure environment
- VPC exists at the region level



Subnet

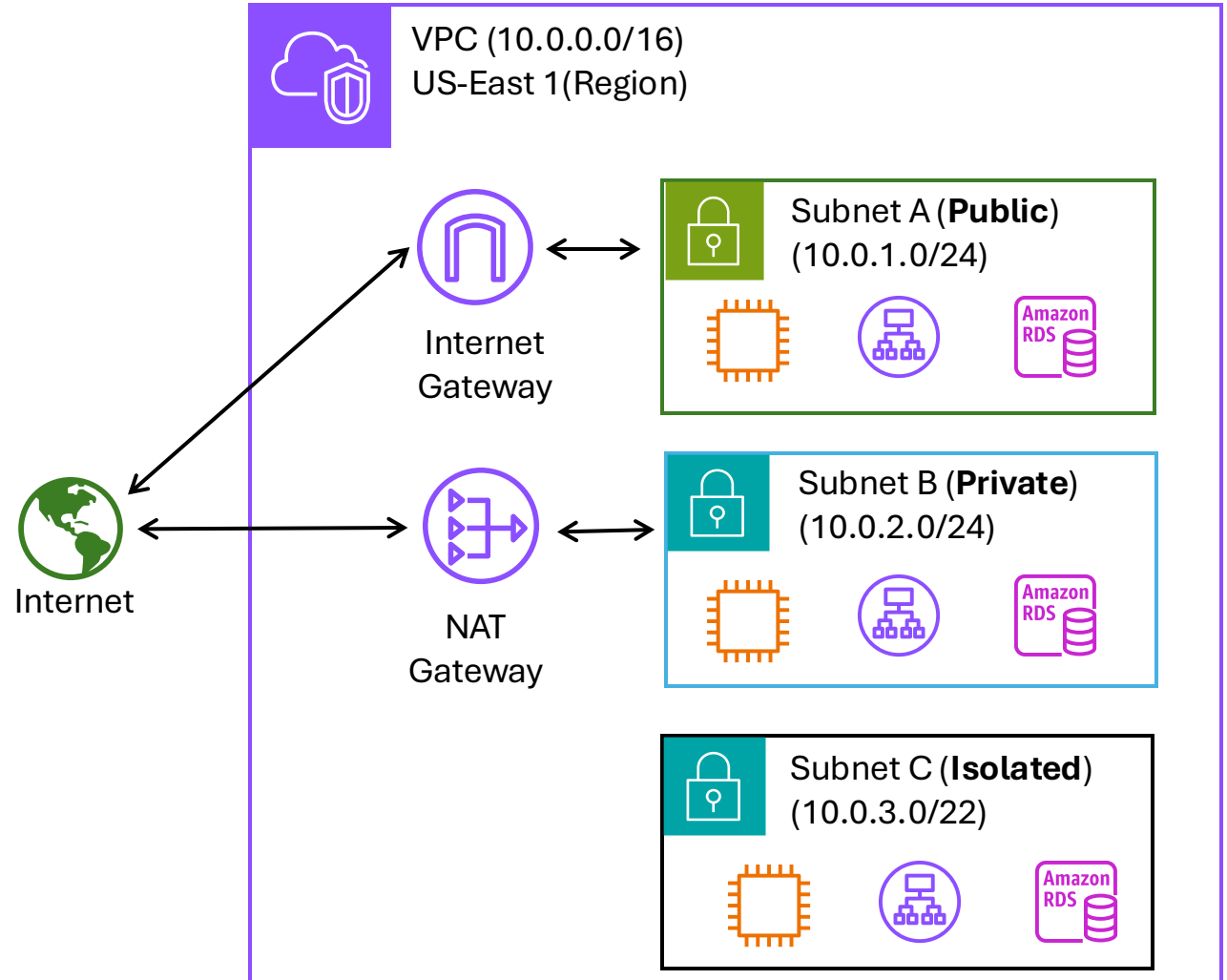
- Subdivision network of a VPC
- Public, private, or isolated
- Resources are launched inside subnets
- Subnets exist within an Availability Zone

	Network	CIDR Block	IP Range	Total IPs
	VPC	10.0.0.0/16	10.0.0.0 – 10.0.255.255	65,534
1st division	Subnet A	10.0.1.0/24	10.0.1.0 – 10.0.1.255	254
2nd division	Subnet B	10.0.2.0/24	10.0.2.0 – 10.0.2.255	254
3rd division	Subnet C	10.0.3.0/22	10.0.3.0 – 10.0.6.255	1,022



Internet Gateway and NAT Gateway

- Internet Gateway
 - Used by public subnets to access the internet
 - Resources need a public IP
- NAT Gateway
 - Used by private subnets to access the internet securely
 - Resources can access internet with private IP

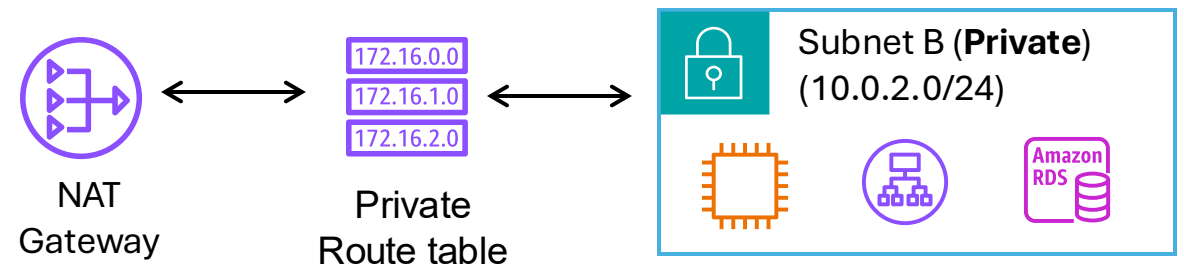
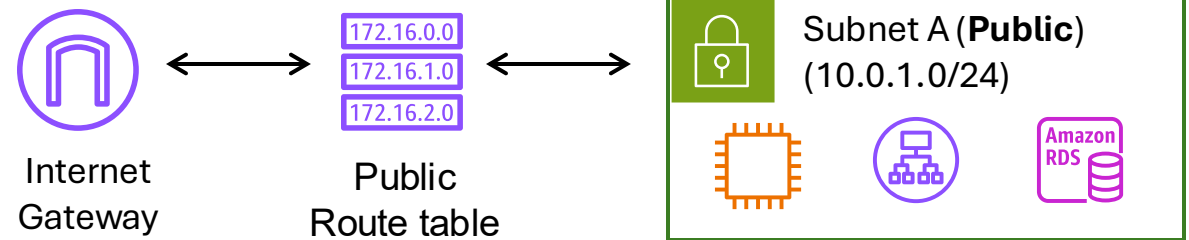


Route Table

- Subnet don't know their route
- Route table defines traffic paths
- Public subnets → Internet Gateway
- Private subnets → NAT Gateway

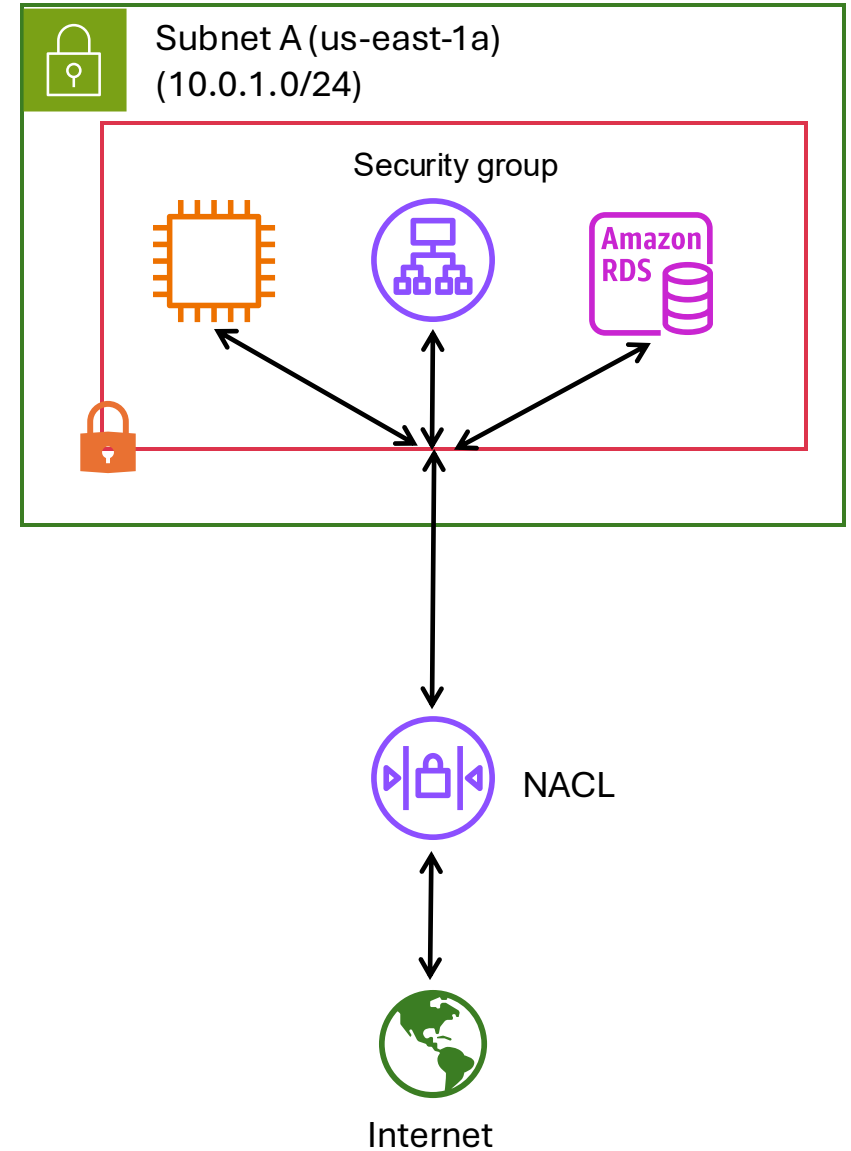
Public Route Table	
Destination	Target
0.0.0.0	Internet Gateway

Private Route Table	
Destination	Target
0.0.0.0	NAT Gateway



NACL and Security Group

- Security Group(SG)
 - Firewall at resource level
 - Controls inbound and outbound traffic
- Network Access Control List(NACL)
 - Firewall at subnet level
 - Rules for inbound and outbound



Thanks for watching!